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Saddle and bridge for reducing longitudinal waves in a string instrument

Abstract

A saddle for a string instrument includes a string contact surface comprising a first material, a saddle end surface, generally opposite the string contact surface, comprising the first material, and two opposing side surfaces comprising a vibration-absorbent material different than the first material.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of co-pending U.S. patent application Ser. No. 16/737,260, filed Jan. 8, 2020, which claims the benefit of U.S. Provisional Patent Application 62/876,217, filed Jul. 19, 2019, the contents of each of which are incorporated herein by reference in their entirety.

BACKGROUND

Field

(1) Embodiments of the present disclosure generally relate to configuration and construction of components of a string instrument. More particularly, the disclosure relates to a saddle and a bridge for reducing longitudinal waves in a string instrument.

Description of the Related Art

(2) A string instrument (sometimes referred to as a stringed instrument) such as a guitar is generally comprised of a solid or hollow resonant body commonly made from one or more woods, or similar material. Attached to this main instrument body is a slender extension commonly referred to as a neck, to which are attached a plurality of strings anchored with adjustable pegs used to control the tension of the strings. The distal end of the strings is attached to a bridge where vibration of the strings is transferred to the body of the instrument in order to amplify the vibration of the strings and make the vibration audible.

(3) The vibrating length of strings is determined by two fixed points of contact perpendicular to the length of the strings, one point near the adjustable anchoring pegs, and one point on the bridge. The strings are stretched taut over these two points of contact. This point of contact on the bridge is typically a saddle comprising hard material for the strings to rest on, often made of natural bone, ivory, or a dense synthetic material and fit tightly into an elongated aperture formed in the hard wood bridge of a guitar. A musician will strum or pluck these strings to set them in motion, creating sound. The pitch of the notes played is determined by stopping the strings against the neck, altering their speaking or vibrating length and corresponding frequency.

(4) When the string of such an instrument, like a guitar vibrates, its motion can be described as the sum of two waveforms, referred to by those familiar in the art as the transverse wave motion and the longitudinal wave motion. The transverse wave motion is characterized by movement of the vibrating string in a direction perpendicular or transverse to the axis of the string when it is at rest. The longitudinal wave motion travels parallel to the axis of the string. On a guitar or other string instrument, the transverse wave is the motion primarily responsible for the audible musical pitch. The frequency of the transverse string motion can be intentionally tuned by altering the tension of the string, as well as the active speaking length. The longitudinal wave typically travels at a higher speed and frequency than the transverse wave, and is more difficult to tune as it's pitch or

frequency cannot be significantly altered by tension. It can be tuned by altering the composition of the string itself to change the material's density or flexibility, or by altering the overall length of the string.

(5) A challenge to overcome in building a string instrument is to balance the transverse and longitudinal motions via string length, size, weight, stiffness, tension and pitch in order to prevent the two vibratory motions from causing interference with each other and corrupting the harmonic sound of the desired musical note.

(6) When the instrument is fitted with an electromechanical pickup sensor, longitudinal wave motion is particularly significant and detrimental to musical functioning of the instrument. Piezo electric crystals are often employed in amplifying such a string instrument. These crystals are extremely sensitive to vibration and respond to vibratory motion of the saddle piece installed in the bridge. When installed in the bridge of a string instrument, an electromechanical pickup system is particularly sensitive to reception of a string's longitudinal wave motion which causes undesirable resonant frequencies and harmonic corruption of the musical frequencies imparted by the transverse wave motion.

(7) Existing techniques for balancing longitudinal waves with transverse waves involve altering the composition and/or length of strings. One method is taught by Harold Conklin (U.S. Pat. No. 3,523,480A), where the active vibrating length of a piano string is fixed so that the transverse wave motion and longitudinal wave motion have frequencies which relate to each other in a predetermined musically pleasing harmonic relationship.

(8) Another existing method is taught by James Ellis (U.S. Pat. No. 5,874,685A), wherein the longitudinal wave form and transverse wave form are determined by altering the string composition or articulation point from a piano hammer or harpsichord plectrum so the resonant frequency of the longitudinal wave is interfered upon by the transverse wave and cancelled out.

(9) However, these existing techniques cannot be employed with guitars. Unlike a piano that uses one or more individual strings to play each note, a guitar is expected to play many notes on each string by altering the length of the transversely vibrating string portion as the player depresses the strings to the frets, continuously altering the relationship between the longitudinal and transverse string vibrations and preventing the use of previously taught methods. Therefore, there is a need in the art for techniques for reducing the audible effect of the longitudinal wave form in guitars and other fretted string instruments

SUMMARY

(10) The present disclosure generally relates to a string instrument, more particularly, components of a guitar.

(11) One embodiment provides a saddle for a string instrument, comprising: a string contact surface comprising a first material; a saddle end surface, generally opposite the string contact surface, comprising the first material; and two opposing side surfaces comprising a vibration-absorbent material different than the first material.

(12) Another embodiment provides a guitar, comprising: a neck; a body; a top; a bridge affixed to the top, the bridge comprising a slot, the slot having a slot end surface and two side walls; and a saddle at least partially disposed within the slot, the saddle having a string contact surface, a saddle end surface generally opposite the string contact surface, and two opposing side surfaces comprising a vibration-absorbent material.

(13) Another embodiment provides a bridge comprising a slot, the slot comprising: a slot end surface; and two side walls, wherein the two side walls comprise a vibration-absorbent material, and wherein a saddle is at least partially disposed within the slot, the saddle having a string contact surface, a saddle end surface generally opposite the string contact surface, and two opposing side surfaces in contact with the two side walls.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.
- (2) FIG. 1A-1G are various views of a prior art saddle.
- (3) FIGS. 2A-2G are various views of a saddle according to embodiments of the present disclosure.
- (4) FIG. 3A illustrates a bridge in which embodiments of the present disclosure may be implemented.
- (5) FIG. 3B illustrates a saddle according to embodiments of the present disclosure disposed within a bridge.
- (6) FIGS. 4A-4G are various views of a saddle according to alternative embodiments of the present disclosure.
- (7) FIG. 5A illustrates a bridge associated with a pickup system in which embodiments of the present disclosure may be implemented.
- (8) FIG. 5B illustrates a saddle according to embodiments of the present disclosure disposed within a bridge associated with a pickup system.
- (9) FIG. 6 illustrates a saddle according to embodiments of the present disclosure disposed within a bridge.
- (10) FIG. 7A illustrates a bridge according to embodiments of the present disclosure.
- (11) FIG. 7B illustrates a saddle disposed within a bridge according to embodiments of the present disclosure.
- (12) FIG. 8 illustrates a guitar in which embodiments of the present disclosure may be implemented.

DETAILED DESCRIPTION

- (13) The present disclosure relates to a saddle and a bridge for reducing longitudinal waves in a string instrument.
- (14) Embodiments of the present disclosure include a modified saddle piece that, when inserted into the bridge atop of which the strings rest on a string instrument, dampens longitudinal waves to prevent them from interfering with the desirable transverse wave motion. Alternative embodiments of the present disclosure include a modified bridge, into which a saddle piece is inserted, that dampens longitudinal waves.
- (15) FIGS. 1A-1G depict different views of a prior art saddle **100** for a string instrument. FIG. 1A is a bottom view, FIG. 1B is an isometric view, FIG. 1C is a side view, FIG. 1D is another side view, FIG. 1E is another side view, FIG. 1F is another side view, and FIG. 1G is a top view of saddle **100**.
- (16) As shown, saddle **100** includes a string contact surface **102** on which strings typically rest. Saddle **100** also includes a saddle end surface **104** which generally contacts the bottom of the slot on the bridge into which saddle **100** is inserted. Saddle **100** also includes two opposing side surfaces **106** and **108**, which generally contact side walls of the slot on the bridge into which saddle **100** is inserted. Saddle **100** also includes two additional side surfaces **152** and **154**, which generally contact additional side walls of the slot of the bridge into which saddle **100** is inserted.
- (17) Saddle **100** is typically made of a hard material such as natural bone, ivory, or a dense synthetic material, and is fit tightly into the slot of a bridge. Vibration of the strings of a string instrument is typically transferred to the body of the instrument through the bridge via saddle **100**

in order to amplify the vibration of the strings and make the vibration audible. However, with prior art saddle **100**, the undesirable longitudinal wave is transferred along with the desirable transverse wave to the body of the instrument.

(18) FIGS. 2A-2G illustrate various views of a saddle **200** for reducing longitudinal waves in a string instrument according to embodiments of the present disclosure. FIG. 2A is a bottom view, FIG. 2B is an isometric view, FIG. 2C is a side view, FIG. 2D is another side view, FIG. 2E is another side view, FIG. 2F is another side view, and FIG. 2G is a top view of saddle **200**.

(19) Similarly to saddle **100** of FIG. 1, saddle **200** includes a string contact surface **202** on which strings typically rest. Saddle **200** also includes a saddle end surface **204** which generally contacts the bottom of the slot on the bridge into which saddle **200** is inserted. Saddle **200** also includes two opposing side surfaces **206** and **208**, which generally contact side walls of the slot on the bridge into which saddle **200** is inserted. Saddle **200** also includes two additional side surfaces **252** and **254**, which generally contact additional side walls of the slot of the bridge into which saddle **200** is inserted. Saddle **200** serves as an end stop for the active speaking length of the strings of a string instrument.

(20) Like saddle **100**, saddle **200** is generally made out of a hard, dense material such as natural bone, ivory, or a dense synthetic material. Unlike saddle **100**, however, saddle **200** has been modified to include a vibration-absorbent material in portions **210**, **220**, **230**, and **240** of its side surfaces **206**, **208**, **252**, and **254**. As used herein, a vibration-absorbent material may comprise rubber, silicone, foam, plastic, or another type of vibration-absorbent material. More generally, the vibration-absorbent material has a lower density than the material from which the rest of saddle **200** is made.

(21) The vibration-absorbent material may be added to saddle **200** in a variety of ways. In some embodiments, portions **210**, **220**, **230**, and **240** of respective surfaces **206**, **208**, **252**, and **254** have been cut or milled away where they would come in contact with the side wall of the saddle slot, and have been filled in or over molded with the vibration-absorbent material. The outer surface of the vibration-absorbent material in portions **210**, **220**, **230**, and **240** is generally flush with the outer surface of the hard material of the rest of side surfaces **206**, **208**, **252**, and **254** of saddle **200**. In alternative embodiments, the vibration-absorbent material may be overlaid onto portions **210**, **220**, **230**, and **240** without cutting or milling away any of the original hard material of saddle **200**. In some embodiments, the vibration-absorbent material extends continuously around the perimeter of saddle **200** to cover portions **206**, **208**, **252**, and **254**.

(22) When saddle **200** is inserted into the slot of a bridge of a string instrument, the vibration-absorbent material in portions **220** serves to dampen longitudinal waves produced by strings while allowing transverse waves to transfer to the body of the string instrument via saddle end surface **204**, which does not include the vibration-absorbent material.

(23) FIG. 3A depicts a bridge **300** of a string instrument. Bridge **300** is generally made out of a hard wood, though may alternatively be made out of other materials that vibrate sympathetically with strings, such as metal or plastic.

(24) Bridge **300** has a slot **310** that is designed for a saddle. A saddle is typically fit tightly into slot **310** so that vibrations from strings are transferred from the saddle to bridge **300**. Bridge **300** is generally attached to a string instrument, and vibrations are transferred from bridge **300** to the body of the string instrument. In some embodiments, as described below with respect to FIGS. 5A and 5B, bridge **300** may be equipped with a pickup.

(25) FIG. 3B illustrates a saddle **200** according to embodiments of the present disclosure disposed within a bridge **300**. For example, saddle **200** may be saddle **200** of FIG. 2, and bridge **300** may be bridge **300** of FIG. 3A.

(26) Saddle **200** is fit tightly into slot **310** of bridge **300**. The bottom surface or saddle end surface **204** of FIG. 2A of saddle **200** rests on a floor of slot **310**, and does not include the vibration-absorbent material, thereby maintaining direct contact between the dense saddle material and the

hard surface of the bridge. Portions **210**, **220**, **230**, and **240** of FIGS. 2B, 2C, 2D, 2E, and 2F of saddle **200** are in contact with side walls of slot **310**. In certain embodiments, saddle **200** is fit within slot **310** such that portions **210**, **220**, **230**, and **240** of FIGS. 2A, 2B, 2E, and 2F extend at least a small amount above the top edge of slot **310**. As such, the vibration-absorbent material covers all portions of the side surfaces of saddle **200** that contact the side walls of slot **310**. String contact surface **202** of FIG. 2C of saddle **200**, on which strings of a string instrument generally rest, protrudes upward from slot **310**.

(27) With saddle **200** and bridge **300** coupled in this manner, the transverse motion of a string is readily transferred to the top of the string instrument unimpeded via the floor of slot **310**. However, the vibration-absorbent material of the side surfaces of saddle **200** serves to absorb and dampen the undesirable longitudinal wave motion, as well as other undesirable high frequency vibration that can interfere with the acoustic sound of the instrument. As such, the use of saddle **200** improves the sound of a string instrument into which it is placed.

(28) In particular embodiments, bridge **300** is equipped with transducers, such as piezoelectric transducers, on the floor of the slot **310**. As such, saddle end surface **204** of FIG. 2A of saddle **200** may rest on top of the transducers. In these embodiments, the vibration-absorbent material of the side surfaces of saddle **200** serves to dampen undesirable high frequency vibration such as longitudinal wave motion, while allowing desirable vibration, such as the transverse motion of the string, to transfer to the transducers via saddle end surface **204** of FIG. 2A.

(29) FIGS. 4A-4G illustrate various views of another saddle **400** for reducing longitudinal waves in a string instrument according to embodiments of the present disclosure. FIG. 4A is a bottom view, FIG. 4B is an isometric view, FIG. 4C is a side view, FIG. 4D is another side view, FIG. 4E is another side view, FIG. 4F is another side view, and FIG. 4G is a top view of saddle **400**.

(30) Similarly to saddle **200** of FIGS. 2A-2F, saddle **400** includes a string contact surface **402** on which strings typically rest. Saddle **400** also includes a saddle end surface **404** which generally contacts the bottom of the slot on the bridge into which saddle **400** is inserted. Saddle **400** also includes two opposing side surfaces **406** and **408**, which generally contact side walls of the slot on the bridge into which saddle **400** is inserted. Saddle **400** also includes two additional side surfaces **452** and **454**, which generally contact additional side walls of the slot of the bridge into which saddle **400** is inserted.

(31) Like saddle **200**, saddle **400** is generally made out of a hard, dense material that has been modified to include a vibration-absorbent material in portions **410**, **420**, **430**, and **440** of its side surfaces **406**, **408**, **452**, and **454**. Unlike saddle **200**, however, portion **420** of saddle **400** does not extend across the entire length of side surface **408**. Rather, portion **420** is interrupted by sections of the original hard material of side surface **408** that have not been modified to include the vibration-absorbent material. In particular, portion **420** is interrupted by three sections of side surface **408** that do not include the vibration-absorbent material. This configuration of side surface **408** is designed to accommodate a pickup. For example, side surface **408** may face the pins that attach strings to a bridge, and the bridge may be equipped with an electromechanical pickup with three sensors, such as piezo crystals. The sensors may contact the sections of side surface **408** that do not include the vibration-absorbent material such that the transverse motion of the strings is transferred to the sensors unimpeded, as described in more detail below with respect to FIGS. 5A and 5B.

(32) FIG. 5A depicts a bridge **500** of a string instrument. Like bridge **300** of FIGS. 3A and 3B, bridge **500** is generally made out of a hard wood, though may alternatively be made out of other materials that vibrate sympathetically with strings, such as metal or plastic, or other materials that allow string vibration to transfer through to the body of the guitar.

(33) Bridge **500** has a slot **510** that is designed for a saddle. Bridge **500** is generally attached to a string instrument, and vibrations are transferred from bridge **500** to the body of the string instrument. Bridge **500** also includes an electromechanical pickup with three sensors **520**. Sensors **520** may be transducers, such as piezoelectric transducers. For example, sensors **520** may be part of

a pickup assembly for receiving vibrations and converting them to electric signals in order to amplify or record the sound made by strings. In some embodiments, a vibration-absorbent material is included behind sensors **520** in bridge **500**.

(34) FIG. **5B** illustrates a saddle **400** according to embodiments of the present disclosure disposed within a bridge **500**. For example, saddle **400** may be saddle **400** of FIG. **4**, and bridge **500** may be bridge **500** of FIG. **5A**.

(35) Saddle **400** is fit tightly into slot **510** of bridge **500**. The bottom surface or saddle end surface **404** of FIG. **4D** of saddle **400** rests on a floor of slot **510**, and does not include the vibration-absorbent material, thereby maintaining direct contact between the dense saddle material and the hard surface of the bridge. Portions **410**, **420**, **430**, and **440** of FIGS. **4B**, **4C**, **4D**, **4E**, and **4F** of saddle **400** are in contact with side walls of slot **510**. In certain embodiments, saddle **400** is fit within slot **510** such that portions **410**, **420**, **430**, and **440** of FIGS. **4B**, **4C**, **4D**, **4E**, and **4F** extend at least a small amount above the top edge of slot **510**. As such, the vibration-absorbent material covers all portions of the side surfaces of saddle **400** that contact the side walls of slot **510**.

(36) Side surface **408** of FIG. **4B** of saddle **400** is positioned so that the sections that do not include the vibration-absorbent material, the sections that interrupt portion **420**, are in contact with sensors **520**. As such, the hard surface of saddle **400** is placed in contact with sensors **520** in order to transfer the transverse waves from the strings to sensors **520**, while the rest of the side surfaces of saddle **400** that contact the side walls of slot **510** are covered in the vibration-absorbent material in order to dampen the longitudinal waves.

(37) String contact surface **402** of FIG. **4C** of saddle **400**, on which strings of a string instrument generally rest, protrudes upward from slot **410**.

(38) With saddle **200** and bridge **300** coupled in this manner, the transverse motion of a string is readily transferred to the top of the string instrument unimpeded via the floor of slot **310** and to sensors **520** via the sections of side surface **408** that do not include the vibration-absorbent material. However, the vibration-absorbent material of portions **410**, **420**, **430**, and **440** of FIGS. **4B**, **4C**, **4D**, **4E**, and **4F** serves to absorb and dampen the undesirable longitudinal wave motion, which can interfere with the acoustic sound of the instrument, as well as the sound signal when the instrument is fitted with an electromechanical pickup system including sensors **520**. As such, the use of saddle **400** improves the sound of a string instrument into which it is placed, both unplugged and through a pickup.

(39) FIG. **6** illustrates a saddle **650** according to embodiments of the present disclosure disposed within a slot of a bridge **600**. Saddle **650** may be representative of either saddle **200** of FIGS. **2A-2F** or saddle **400** of FIGS. **4A-4F**. Bridge **600** may be representative of either bridge **300** of FIGS. **3A-3B** or bridge **500** of FIGS. **5A-5B**.

(40) Saddle **600** has a side surface **608** including a portion **610** that comprises a vibration-absorbent material. As illustrated, portion **610** extends a small amount above the surface of bridge **600**, thereby ensuring that no part of saddle **650** not covered in the vibration-absorbent material is in contact with the side walls of the slot in bridge **600** into which saddle **650** is inserted.

(41) FIG. **7A** illustrates a bridge **700** according to embodiments of the present disclosure.

(42) Bridge **700** is generally made of a hard material and includes a slot **710**, similarly to bridge **300** of FIGS. **3A-3B** and bridge **500** of FIGS. **5A-5B**. However, on bridge **700**, the side walls **720** of slot **710** are covered in a vibration-absorbent material. For example, the hard material on the side walls **720** of slot **710** may have been cut or milled down and filled in or over molded with the vibration-absorbent material. When a saddle is inserted tightly into slot **710** of bridge **700**, the vibration-absorbent material on the side walls **720** of slot **710** serves to dampen the longitudinal waves produced by strings resting on the saddle, while still allowing the transverse waves to transfer to the body of the instrument via a floor of slot **710**. In alternative embodiments, the vibration-absorbent material may be added to side walls **720** without cutting or milling any portion of side walls **720**. In these embodiments, a smaller saddle may be inserted into slot **710**.

(43) Furthermore, in some embodiments, bridge **700** includes a pickup system comprising sensors, such as piezoelectric transducers. As such, the vibration-absorbent material may only cover the portions of side walls **710** that do not include sensors.

(44) FIG. **7B** illustrates a saddle **750** disposed within a bridge **700** according to embodiments of the present disclosure. For example, saddle **750** may be representative of prior art saddle **100** of FIGS. **1A-1F** and saddle **700** may be saddle **700** of FIG. **7A**.

(45) Saddle **750** is fit tightly into slot **710** of bridge **700**. The side surfaces of saddle **700** contact the vibration-absorbent material on side walls **720** of slot **710**.

(46) With saddle **750** and bridge **700** coupled in this manner, the transverse motion of a string is readily transferred to the top of the string instrument unimpeded via the floor of slot **710** (and, in some embodiments, to sensors of a pickup system). However, the vibration-absorbent material of side walls **720** serves to absorb and dampen the undesirable longitudinal wave motion, which can interfere with the acoustic sound of the instrument, as well as the sound signal when the instrument is fitted with an electromechanical pickup system. As such, the use of bridge **700** improves the sound of a string instrument into which it is placed, both unplugged and through a pickup.

(47) FIG. **8** depicts a guitar **800** with which embodiments of the present disclosure may be implemented.

(48) In the example of FIG. **8**, the guitar is an acoustic guitar wherein the top of the guitar acts as an acoustic soundboard, but elements of the present invention are equally useful when applied to an electric guitar or any other string instrument. The guitar includes a body **810**, a neck **820**, and a headstock **830**. Strings, including string **825**, extend from the headstock where they are tightened to a preferred tension with keys **840** to a bridge **850** (e.g., bridge **300** of FIGS. **3A-3B**, bridge **500** of FIGS. **5A-5B**, or bridge **700** of FIGS. **7A-7B**) where they are anchored with bridge pins **855**, one for each string. A nut **860** is placed at the end of a fingerboard **865** adjacent the headstock and controls the string spacing, distance from the edge of the fingerboard and the height of the strings above a first fret **870** on the fingerboard **865**. The strings are slightly splayed over their length and extend over a saddle **875** that is housed in the bridge **850**. Saddle **875** may be saddle **100** of FIGS. **1A-1F**, saddle **200** of FIGS. **2A-2F**, or saddle **400** of FIGS. **4A-4F**. The portion of the strings that vibrates to create a sound when plucked is that portion extending between the nut **860** and saddle **875**. The strings are stopped or effectively shortened when they are depressed behind a fret.

(49) It is noted that, while certain embodiments are described with respect to guitars, techniques presented herein may also be employed with other types of string instruments. While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A saddle for a string instrument, comprising: a string contact surface comprising a first material; a saddle end surface, generally opposite the string contact surface, comprising the first material; and two opposing side surfaces comprising a vibration-absorbent material different than the first material, wherein the vibration-absorbent material is disposed within a depression in the first material on an outer portion of at least one side surface of the two opposing side surfaces such that the vibration-absorbent material does not extend beyond the saddle end surface.
2. The saddle of claim 1, wherein a first side surface of the two opposing side surfaces comprises: at least a first section comprising the first material; and a plurality of second sections comprising the vibration-absorbent material.
3. The saddle of claim 2, wherein two second sections of the plurality of second sections are separated by the first section.
4. The saddle of claim 2, wherein the first side surface is a pin side of the saddle.

5. The saddle of claim 1, wherein an outer surface of the vibration-absorbent material is generally flush, on the outer portion of the least one side surface, with an outer surface of the first material.
 6. The saddle of claim 1, wherein the vibration-absorbent material is selected from the following list: rubber; silicon; plastic; or foam.
 7. The saddle of claim 1, wherein the vibration-absorbent material has a lower density than the first material.
 8. The saddle of claim 1, further comprising two additional opposing side surfaces, generally perpendicular to the two opposing side surfaces, comprising the vibration-absorbent material.
 9. The saddle of claim 8, wherein the vibration-absorbent material extends continuously around the two opposing side surfaces and the two additional opposing side surfaces.
 10. A guitar, comprising: a neck; a body; a top; a bridge affixed to the top, the bridge comprising a slot, the slot having a slot end surface and two side walls; and a saddle at least partially disposed within the slot, the saddle having: a string contact surface comprising a first material; a saddle end surface, generally opposite the string contact surface, comprising the first material; and two opposing side surfaces comprising a vibration-absorbent material, wherein the vibration-absorbent material is disposed within a depression in the first material on an outer portion of at least one side surface of the two opposing side surfaces such that the vibration-absorbent material does not extend beyond the saddle end surface.
 11. The guitar of claim 10, further comprising at least a first transducer located on a side wall of the slot, the first transducer having a transducer contact surface in contact with a section of a side surface of the two opposing side surfaces of the saddle, wherein the section of the side surface comprises a material other than the vibration-absorbent material.
 12. The guitar of claim 10, wherein a first side surface of the two opposing side surfaces comprises: at least a first section comprising the first material; and a plurality of second sections comprising the vibration-absorbent material.
 13. The guitar of claim 12, wherein two second sections of the plurality of second sections are separated by the first section.
 14. The guitar of claim 12, wherein the first side surface is a pin side of the saddle.
 15. The guitar of claim 10, wherein an outer surface of the vibration-absorbent material is generally flush, on the outer portion of the at least one side surface, with an outer surface of the first material.
 16. The guitar of claim 10, wherein the vibration-absorbent material is selected from the following list: rubber; silicon; plastic; or foam.
 17. The guitar of claim 10, wherein the vibration-absorbent material has a lower density than the first material.
 18. A bridge comprising a slot, the slot comprising: a slot end surface comprising a first material; and two side walls, wherein the two side walls comprise a vibration-absorbent material different than the first material, wherein the vibration-absorbent material is disposed within a depression in the first material on at least one side wall of the two side walls, wherein a saddle is at least partially disposed within the slot, the saddle having a string contact surface, a saddle end surface generally opposite the string contact surface, and two opposing side surfaces in contact with the two side walls, and wherein the vibration-absorbent material does not extend beyond the saddle end surface.
 19. The bridge of claim 18, wherein the vibration-absorbent material is selected from the following list: rubber; silicon; plastic; or foam.
 20. The bridge of claim 18, wherein the vibration-absorbent material has a lower density than the first material.
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